

ORIGINAL COMMUNICATION

Rheumatoid Arthritis in Review: Clinical, Anatomical, Cellular and Molecular Points of View

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Rheumatoid arthritis (RA) is the most common chronic autoimmune disease of the joints affecting close to 0.5–1.0% of the general population. Although the etiopathogenesis of RA remains elusive, the involvement of dendritic cells and type 17 T-helper cells appears to be pivotal in maintaining a state of chronic inflammation. RA is generally characterized by small joint involvement. A chronic inflammatory process leads to joint destruction and to tendon and ligament laxity and disintegration. These processes result in an imbalance of forces acting on the joints causing joint deformities including swan neck deformity, boutonniere deformity of the hands, flexion deformity of the wrist, lesser toe deformities, and others. In some instances, bony erosions subsequent to the RA disease process can result in life-threatening events including, for example, atlanto-axial subluxation, which can cause myelopathy and paralysis; and basilar invagination, which can cause brain stem injury and imminent death. Although less commonly involved, larger joints are not spared, as evidenced by the involvement of the elbow, hip, and shoulder joints in a sizable proportion of RA patients. The progression and prognosis of this disease entity are variable, guarded and dependent on the efficacy and response to treatment modalities employed. Inadequate management results in disease progression, which ultimately leads to joint erosion, destruction, deformities and substantial decrease in the functional quality of life. Clin. Anat. 31:216–223, 2018. © 2017 Wiley Periodicals, Inc.

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INTRODUCTION

Rheumatoid arthritis (RA) is the most common chronic autoimmune disease of the synovial joints. Ultimately it causes persistent inflammation, which results in joint swelling, deformity, and degradation of daily functional capacity and quality of life. Although RA can essentially affect all synovial lined joints, it has a predilection for the small joints of the hands and feet before it affects larger joints including the knees, shoulders, hips and others (Suresh, 2004). Generally, RA affects the joint in a polyarticular and symmetrical pattern, though there are atypical presentations in a subset of patients (Suresh, 2004).

Although RA is characterized by joint involvement, it also presents with a constellation of manifestations unrelated to locomotion, known as extra-articular manifestations (Cojocaru et al., 2010). These include rheumatoid nodules, vasculitis, pleural effusion, accelerated atherosclerosis, and others (Cojocaru et al.,

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2010). They can occur in 40% of RA patients, either with disease onset or subsequently (Cimmino et al., 2000).

Epidemiological studies demonstrate that RA affects approximately 0.5–1% of the general population. Its prevalence and incidence are regionally variable. Existing data show the highest incidence rates of RA in Pima Indians (5.3%) and Chippewa Indians (6.8%). In contrast, the incidence is lowest amongst the Chinese and Japanese (0.2–0.3%) (Silman and Pearson, 2002). It has been proposed that this regional variability in disease epidemiology is a consequence of distinctive patterns of genetic makeup and differences in exposure to environmental factors. It has also been recognized that women are at a greater risk of disease development than men, in a 2:1 ratio; the reason remains to be elucidated (Turesson and Matteson, 2004). Although progress has been made in clarifying the mechanisms involved in RA, our understanding of the specific cause that initiates the cascade of RA remains incomplete. Leading hypotheses suggest the important role of exposure to environmental triggers in a genetically susceptible host, such as infections by Epstein-Barr virus, parvovirus and mycoplasma. Although the mechanism of disease initiation remains elusive, molecular mimicry has been implicated (Li et al., 2013). Moreover, cigarette smoking has been shown not only to increase the risk of RA in an independent fashion but also to influence the development of positive rheumatoid factor (RF) (McInnes and Schett, 2011). Inter-twin genetic studies showed a concordance rate of around 30% between monozygotic twins, highlighting the role of genetic factors in disease etiopathogenesis (MacGregor et al., 2000). RA susceptibility has also been linked to genetic differences in the human leukocyte antigen (HLA)-DRB1 allele (Gregersen et al., 1987). Other possible explanations for the disease etiopathogenesis include the influence of the gut/oral microbiome and the effect of epigenetic modification, which eventually leads to loss of tolerance and ultimately causes synovitis and overt arthritis (Smolen et al., 2016).

The outcome and prognosis of RA is guarded and depends on several factors including the nature of disease progression, prompt diagnosis, and the response to disease-modifying therapies (Heidari, 2011). Clarification of the immune-pathogenic mechanisms that orchestrate the state of chronic inflammation associated with RA fostered the development of targeted agents including biological therapies that revolutionized the standards of care for RA patients. Early diagnosis and effective treatment have been clearly shown to induce remission and prevent disease progression and deformity development (Heidari, 2011).

The purpose of this article is to review the immunological and anatomical changes characteristic of RA and to provide insight into the debilitating nature of this disease entity.

ANATOMY OF SYNOVIAL JOINT AND PERIARTICULAR BONE

The synovial joint is the most common type of joint in the body and is characterized by the synovial cavity. The articular capsule, a fibrous connective tissue,

attaches to the periarticular bones and encloses the synovial cavity within it (Walmsley, 1961). The bones in the synovial joint, in contrast to other joint types, are protected from direct contact by the articular cartilage, a highly specialized hyaline connective tissue that provides a lubricating surface for efficient articulation to facilitate free locomotion (Walmsley, 1961).

The joint capsule, which is continuous with the periosteum of the periarticular bones, is composed mainly of two layers. The outer fibrous connective tissue is formed by interwoven bundles of collagen fibers that encapsulate the joint. This fibro-cartilaginous layer also has an integral role in joint locomotion and stability. The joint capsule is locally thickened to form capsular ligaments, while in other instances the capsule provides attachment points for the tendons (Ralphs and Benjamin, 1994). Structurally, the attachment zone of the articular capsule and the bone can be divided into four regions: pure fibrous tissue, non-calcified fibrocartilage, calcified fibrocartilage, and bone (Apostolakis et al., 2014). Histologically, the attachment zone is composed of Type II collagen, which is characteristic of cartilage (Ralphs et al., 1992). It has been documented that joint aging leads to the spread of Type II collagen from the attachment zone to nearby tendons and ligaments. Alterations in the attachment zone have also been documented in disease entities such as RA (Benjamin et al., 1993; Sophia Fox et al., 2009).

On the inner surface of the articular capsule lies the second layer, the thin synovial membrane, the function of which is to secrete synovial fluid, a highly viscous straw-colored fluid that has several functions such as joint lubrication and to serve as a medium for the passage of nutrients (Tercic and Bozic, 2001). The synovium in an otherwise healthy patient has a unique cellular lining. The synovial cells lining the synovium are called synoviocytes and are responsible for the production and resorption of synovial fluid and exchange with the blood (Iwanaga et al., 2000). Type A synoviocytes are derived from the monocyte/macrophage cell lineage. In contrast, Type B synoviocytes are of mesenchymal origin and are referred to as fibroblast-like synoviocytes. Type A synoviocytes are actively involved in the phagocytosis of waste and cell debris in the joint cavity. They also have antigen-presenting capability. Type B synoviocytes are characterized by a high density of rough endoplasmic reticulum and are believed to have an integral role in specialized matrix production and synovial fluid synthesis (Iwanaga et al., 2000).

Finally, periarticular bone is usually composed of two bone types: trabecular bone, which makes the inner layer and has a honeycomb structure; and cortical bone, which is dense and compact and forms the outer layer (Clarke, 2008).

IMMUNOPATHOLOGY OF SYNOVITIS IN RA

Despite the clinical relevance and high prevalence of RA, the mechanisms involving disease pathogenesis are still elusive. Accumulated evidence has shown that dendritic cells are crucial for its initiation and

for the perturbation of RA-associated synovitis (Rodriguez-Fernandez, 2013). These dendritic cells constitute up to 5% of synovial fluid mononuclear cells. Generally, those present in the synovial fluid express HLA Class II molecules and co-stimulatory molecules (CD80/86), both of which are essential for antigen presentation and T cell activation (Huppa and Davis, 2003; McInnes and Schett, 2011). When activated, dendritic cells increase the release of interleukin-1 (IL-1), IL-6, IL-23, tumor necrosis factor- α (TNF- α), and other factors. The activation of dendritic cells results in the stimulation and selective differentiation of specific T-cell subsets (McInnes and Schett, 2011). Although Type 1T-helper (Th-1) cell cytokines can be detected in RA synovial tissue, their levels are lower than other cytokines, indicating a marginal role in disease pathogenesis compared to other T-cell subsets (Firestein, 2003). In contrast, both IL-6 and transforming growth factor- β drive the differentiation and polarization of naïve T-cells to Th-17 cells (Annunziato et al., 2007). IL-23, which is a member of the IL-12 family, is secreted from dendritic cells and mediates Th-17 cell lineage expansion and survival (Annunziato et al., 2007). Th-17 is notable for secreting the cytokine IL-17, which effects a wide array of functions by affecting multiple cell lines including monocytes, fibroblasts, osteoclasts and chondrocytes. IL-17 orchestrates chronic inflammation and joint destruction through its influence on cytokine release from those activated cells, including for instance IL-1, IL-6, TNF- α , prostaglandin E2, metalloproteinases, and other molecules (Lundy et al., 2007). Notably, the upregulated expression of the RANK (receptor activator of NF- κ B) ligand on osteoblasts has been shown to follow IL-17 secretion. The upregulation of RANKL promotes the formation of osteoclasts, the agents of bone dissolution (Sato et al., 2006).

Although dendritic cells and Th-17 are pivotal in disease pathogenesis, other cells are also involved in this inflammatory process. Accumulated experimental evidence has demonstrated the possible role of fibroblast-like synoviocytes in antigen presentation following their high expression of MHC II molecules. Additionally, interactions between activated T-cells and fibroblast-like synoviocytes result in the release of pro-inflammatory cytokines (Tran et al., 2005; Zimmermann et al., 2001). B cells also have a role in the immune cascade, as indicated by the presence of autoantibodies against citrullinated peptides (ACPA) and RF. Furthermore, clinical evidence has highlighted the efficacy of B cell-depleting agents in disease control and management (Edwards et al., 2004).

ACPA form immune complexes by binding to self-proteins including but not limited to enolase, fibronectin, histones and Type II collagen (Zhao et al. 2008; Reynisdottir et al., 2016). They also activate macrophages through toll-like receptors, and it was demonstrated that ACPA immune complexes activate osteoclasts and induce bone erosion and osteolysis (Harre et al., 2012).

The joint swelling in RA occurs secondary to joint inflammation, immune activation, and white blood cell infiltration into the synovial compartment. Rheumatoid pannus, which is characteristic of RA, is composed of

these inflammatory cells. The previously mentioned release of proteolytic enzymes results in cartilage matrix destruction, ligament laxity and ultimately joint damage (McInnes and Schett, 2011).

The increased expression of RANKL on T-cells, B-cells and monocytes and its binding to its receptor RANK, which is abundant on macrophages and dendritic cells, stimulates osteoclastogenesis (Geusens, 2012). Moreover, osteoprotegerin, a RANKL receptor antagonist, is downregulated at the mRNA level in RA; and macrophage colony stimulating factor is released from synovial fibroblasts and is important in osteoclast formation (Bottini and Firestein, 2013). Together, all these factors result in bone erosion, which starts at the bare area inside the joint (Geusens, 2012).

THE CONCEPT OF BONE EROSION

RA is generally a diagnosis based on history taking and physical examination. Supporting laboratory tests including elevated acute phase reactants, RF and ACPA can also help in establishing the diagnosis. Another equally important diagnostic modality for RA is radiographic imaging.

RA can be accompanied by radiographic signs including joint space narrowing, erosions, and joint subluxation (Pfeil et al., 2013). Some of these events can occur in recent-onset arthritis while others develop at a much later stage of the RA disease process (Pfeil et al., 2013).

Bone erosions are usually detected by conventional radiographic techniques, and are seen as interruptions in cortical bone, sometimes coexisting with neighboring trabecular bone loss (Schett and Gravallesse, 2012). On the other hand, bone cysts are isolated osteolytic lesions in the trabecular bone without the involvement of cortical bone (Schett and Gravallesse, 2012). Bone erosions are not unique to RA, since several other inflammatory arthritic diseases exhibit patterns of bone erosion including psoriatic arthritis, gout, and even osteoarthritis. Nonetheless, the distinctive distribution pattern of bone erosions, which includes the involvement of multiple symmetrical peripheral small- to medium-sized joints, suggests a diagnosis of RA (Schett and Gravallesse, 2012).

Bone erosions do not occur haphazardly. Instead, a predilection for specific anatomical sites is apparent in RA (Døhn et al., 2007). For example, the radial aspects of small finger joints are highly active in bone destruction and erosion, in contrast to the less-affected ulnar aspects and the virtually spared palmar and volar surfaces of joints (Stach et al., 2010). These lesions appear to prefer the bare area of the joint, mineralized cartilage, or surrounding the insertion points of periarticular ligament and tendons (Schett and Gravallesse, 2012).

Apart from being essential for RA diagnosis, the number and size of bone erosions constitute a key measure of disease severity, progression, and disease response to the therapeutic measures employed (Welsing et al., 2001; Ødegård et al., 2006).

THE ANATOMY OF RHEUMATOID JOINTS

Rheumatoid Hands

In 75% of RA patients involvement of the hand joints and tendons is documented (Ilan and Rettig, 2003). As witnessed in other synovial joints, constant inflammation and pannus formation result in articular cartilage and capsule destruction. These effects interfere with the normal mechanisms involved in hand movement and also weaken the supporting ligament attached to the capsule (Kim and Jung, 2007). The inflamed synovium also infiltrates the tendons surrounding the capsule and can potentially cause their dissolution and in some instances their rupture (Haselwood and Castles, 1977). Therefore, it is expected that tenosynovitis and ligament laxation ultimately result in joint deformities due to muscle imbalance around these joints (Brewerton, 1971; Haselwood and Castles, 1977).

The involvement of the wrist is common in RA producing complaints such as pain, joint swelling and a limitation of motion. Joint swelling can be witnessed on both the dorsal and volar aspects of the wrist in an irregular, multilobular pattern (Brewerton, 1971).

Involvement of the radiocarpal joint results in rotatory instability of the scaphoid bone, which subsequently causes carpal height loss and wrist collapse and therefore ulnar and volar translocation of the carpus on the radius (Rosen and Weiland, 1998). With the advancement of RA, laxity in the ulnar aspect of the joint subsequent to the stretching of restraining ligaments results in dorsal prominence of the ulnar styloid process (Kleinert and Frykman, 1973). A flexion deformity is also characteristic of advanced rheumatoid wrist disease. It occurs as a consequence of radiocarpal joint destruction and the stronger action of the flexor tendons as opposed to the weaker extensor tendons (Rosen and Weiland, 1998; Straub, 1969). As a result, the 30 degree dorsiflexion that represents the normal functional position of the wrist is lost (Straub, 1969).

Involvement of the flexor retinaculum and the compressive effect of synovial pannus potentially result in medial nerve compression (Mitchell et al., 2009). Pain and paresthesia in the innervating course of the median nerve along with characteristic night pain and positive Tinel and Phalen signs can be demonstrated in such patients (Mitchell et al., 2009). In a subset of RA patients, carpal tunnel syndrome can be the initial presenting sign (Phalen, 1966).

The involvement of metacarpophalangeal joints in RA results in hand swelling and limitation of motion. With advancement of RA, a flexion deformity of the metacarpophalangeal joint develops secondary to the involvement of both the lumbricals and interossei of the hand (Currey, 1964). Generally, these two groups of muscles are responsible for metacarpophalangeal joint flexion and interphalangeal joint extension. The involvement of their tendons subsequent to RA results in their fibrosis and spasm, which exaggerates their normal flexion action (Currey, 1964). In addition to the observed increased flexion at the metacarpophalangeal

joints, ulnar deviation of the fingers can potentially occur secondary to many factors including the pull of the lumbricals, the ulnar mobility of the metacarpophalangeal joints, and the ulnar inclination of the metacarpal heads (Currey, 1964).

Interphalangeal joint involvement can present with various deformities. Swan neck deformity usually occurs as a result of proximal interphalangeal joint hyperextension brought about by the contracture of digital intrinsic muscles, and distal interphalangeal joint flexion subsequent to the flexor action of the flexor profundus (Heywood, 1979). Boutonniere deformity, in which the patient exhibits flexion of the proximal interphalangeal joint while hyperextending the distal interphalangeal joint, is believed to result from rupture of the central slip of the extensor tendon (Ferlic, 1989). In contrast, mallet finger occurs subsequent to weakening of the extensor tendon insertion and a resultant incomplete extension of the distal phalanx (Apfelberg et al., 1978). Other deformities include trigger finger. Generally, the rheumatoid thumb can show similar effects including flexion deformity at the metacarpophalangeal joint and hypertension of the interphalangeal joint. Moreover, thumb opposition is usually limited following carpal metacarpal joint involvement (Stein and Terrono, 1996).

Cervical Spine Involvement

RA characterized by erosive synovitis has a tendency to affect the cervical spine; RA is thought to be the most common cause of spine inflammation (Reiter and Boden, 1998). Although it is more likely to affect women than men, men have a greater risk of advanced cervical spine disease (Weissman et al., 1982). The prevalence of cervical disease amongst RA patients ranges from as low as 25% to as high as 80% (Reiter and Boden, 1998). Accumulated evidence has shown that cervical disease begins early in the disease process and its progression is related to the progression of peripheral joint disease (Nguyen et al. 2004). Among RA patients the incidence of neck pain has been reported to be around 40–88%. Cervical subluxation has also been noted in 43–88% of the patients as documented by radiography (Boden, 1994). Although a high percentage of RA patients have radiographic signs, neurological impairment has been documented in only 7–34% (Pellicci et al., 1981). Interestingly, neurological progression occurred in only 36% of RA patients, while radiographic cervical progression was noted in 80% in the same group (Pellicci et al., 1981).

The mechanism that leads to cervical joint synovitis and erosion is similar to the mechanisms discussed previously. The joints most susceptible to the rheumatoid process in the cervical spine are the occiput-C1 and C1-C2 joints (Kim et al., 2015). Their higher susceptibility stems from their unique anatomy: (1) they lack cushioning discs, in contrast to the subaxial cervical spine which possesses discs anteriorly and synovial facets posteriorly that are not prone to the autoimmune process; (2) the C1-C2 joint is amenable to subluxation because the joint articulation is unstable; C1-C2 possess horizontal/axial-oriented articulations

that are prone to subluxation owing to the lack of an interlocking bone to prevent it (Kim et al., 2015). Although the dens on C2 and the transverse cruciate ligament stabilize the joint under normal conditions, injury to them subsequent to RA cancels out their restraining effects (Kim et al., 2015).

Although cervical spine pathologies can coexist, three distinct entities have been identified including the more common atlantoaxial subluxation (65%), followed by basilar invagination (20%), and finally subaxial subluxation (15%) (Lonner et al., 1995).

Atlantoaxial Subluxation

Atlantoaxial subluxation is also known as C1–C2 instability. Subluxation can occur in more than one plane so anterior, posterior, lateral and rotatory subluxations are possible. Most subluxations occur anteriorly with the subluxation of C1 on C2. In 20% of cases there is lateral subluxation, and in 7% a posterior subluxation (Robinson, 1966).

Anterior subluxation usually occurs secondary to the rheumatoid inflammatory process that destroys the transverse, alar and apical ligaments along with bony erosions of the dens. The subluxation process enlarges the anterior atlanto-dens interval and decreases the space available for the spinal cord. Moreover, the presence of rheumatoid pannus around the dens and the anterior aspect of C2 results in further compromise of the space available for the cord, increasing the severity of neurological symptoms due to compressions (Kim et al., 2015).

Atlantoaxial subluxation is diagnosed by cervical spine radiographs. Inspection of flexion and extension views for the anterior atlanto-dens interval could provide insight into the degree of injury; an atlanto-dens interval >5 mm denotes instability, whereas an interval >10 mm means complete loss of structural integrity as a result of ligament disintegration. Although a greater atlanto-dens interval reflects greater severity, correlation with neurological outcome has not been demonstrated (Yang et al., 2014).

Rotatory subluxation, which is far less common, can be appreciated on open-mouth views with significant asymmetry in the width of the lateral masses (Mallory et al., 2014; Kim et al., 2015).

Basilar Invagination

Basilar invagination results from occiput-C1 and C1–C2 erosions that cause a decrease in the vertical distance between the odontoid and the brain stem. This can lead to basilar impingement and possibly severe neurological impairment (Menezes et al., 1985). In view of the serious implications of basilar invaginations, radiographic assessment is essential. McRae's line represents the line between the anterior and posterior median points on the foramen magnum (basion and opisthion, respectively) that the tip of the odontoid is not supposed to cross (Harcourt and Mitchell, 1990). In some instances, involvement of the odontoid in the rheumatoid process decreases the reliability of position judgment. In such instances,

other measurements are employed including Clarke and cephalomedullary angles. Such techniques are no longer employed owing to the availability of more advanced musculoskeletal imaging techniques including MRI and CT (Harcourt and Mitchell, 1990; Sugita et al., 2014).

Subaxial Subluxation

Subaxial subluxation encompasses the various deformities that can occur in the cervical spinal cord on various levels. In some instances it can lead to spondylolisthesis, staircase deformity, or cervical kyphosis (Ito et al., 2009). In RA, as opposed to osteoarthritis, the lack of osteophytes results in further instability, which leads to further deformities and higher degrees of subluxation (Kim et al., 2015).

Neurological Manifestations

Subluxations, erosions, and pannus formation all compromise the space available for the cord. Such events result in the appearance of signs and symptoms that are consistent with myelopathy secondary to cord compression (Mukerji and Todd, 2011). Although some patients can be asymptomatic, others can present with severe symptoms including paralysis. Detailed histories should be taken with focus on problems of ambulation, coordination and bowel movements. Additionally, thorough physical examination with special focus on signs of myelopathy should be conducted including Babinski, Romberg, L'Hermite's signs, clonus and others (Meijers et al., 1984; Mukerji and Todd, 2011).

Foot Joint Involvement

RA patients can present initially with foot and ankle symptoms similar to the joint involvements in the upper extremities. In two distinct reports, 16 and 19% of RA patients initially complained of symptoms related to the foot and the ankle (Vainio, 1956; Fleming et al., 1976). The involvement of foot and ankle joints over the course of RA has been estimated at around 94% (Michelson et al., 1994). RA disease affecting the forefoot has been reported to precede ankle involvement (Fleming et al., 1976; Borman et al., 2012). Involvement of the forefoot and the ankle is also twice as common as involvement of the midfoot and hindfoot (Fuchs et al., 1989; Michelson et al., 1994). Although radiographic changes have been demonstrated in the midfoot, only one third of those patients had clinical symptoms including pain (Vidigal et al., 1975).

RA-associated synovitis and pannus formation results in deformities of the foot. Dorsal dislocation of the metatarsophalangeal joint occurs as a consequence of connective tissue disarrangement (Brooks and Hariharan, 2013). Soft tissue disruption causes metatarsophalangeal joint instability, which is also aggravated by gait and locomotion resulting in increased joint loading and plantar displacement of the metatarsal joints (Brooks and Hariharan, 2013).

Additionally, proximal phalanx dorsal subluxation impedes the normal extensor function of the interosseous muscles, which leads to excessive flexion and subsequent lesser toe deformities including claw toe deformity and hammer toes (Stainsby, 1997). Long-standing and untreated flexion deformities lead to the development of painful calluses over the prominent metatarsal heads (Myerson, 1992; Stainsby, 1997).

Non-articular manifestations subsequent to chronic synovitis include tendonitis (Jaakkola and Mann, 2004). Posterior tibial tendonitis gained special interest because of its proposed role in the development of hindfoot valgus and pesplanus, which are common findings in RA patients (Spiegel and Spiegel, 1982). The role of posterior tibial tendonitis in valgus and pesplanus development has been challenged by the possible role of ligament laxity subsequent to tarsal and subtalar joint inflammation (Keenan et al., 1991). Published work also showed that while 65% of RA patients had pesplanus, only 11% had posteriotibial tendonitis as seen on clinical examination (Michelson et al., 1994).

Rheumatoid nodules are another extra-articular manifestation in the soft tissues and are usually associated with chronic inflammation. They are non-tender masses more likely to occur on extensor surfaces such as the medial and lateral malleoli and medial and lateral eminences of the forefoot, points that correspond to higher pressure areas (O'Brien et al., 1997). Histologically, these lesions are made of granulation tissue (O'Brien et al., 1997).

Additional extra-articular manifestations include neurological manifestations such as tarsal tunnel syndrome and interdigital neuroma. These are believed to occur subsequent to compression caused by rheumatoid nodules, rheumatoid pannus, valgus deformity or other reasons (Vainio, 1979; Ibrahim et al., 2013).

Large Joint Involvement

RA is characterized by small joint involvement, yet in many cases the rheumatoid inflammation involves other joints including large ones. Generally, large joints are expected to become involved in long-term RA, which is associated with a poorer prognosis (Heidari, 2011).

Involvement of the elbow has been documented in 20–65% of RA patients (Studer and Athwal, 2011). Upon examination of the joint, elbow synovitis demonstrated by posterolateral joint swelling can be accompanied by rheumatoid nodules and olecranon bursitis (Sanchez-Sotelo, 2016). Restriction in the ranges of motion can be expected including the early loss of joint extension. Moreover, involvement of the elbow in RA can result in ulnar nerve injury secondary to synovitis (Studer and Athwal, 2011). Radiographic examination reveals a similar finding to that expected in RA patients including osteopenia, joint space narrowing, and bony erosions (Gill and Morrey, 1998; Studer and Athwal, 2011).

Almost 60% of RA patients complain of shoulder pain. Initially, the glenohumeral joint is affected, which results in joint tenderness and swelling

(Thomas et al., 2006; Voorde et al., 2015). In the shoulder, the disease starts by affecting the glenohumeral joint and spreads to involve surrounding structures (Thomas et al., 2006). Entrapment of rotator cuff tendons between the inflammatory pannus inferiorly and the inflamed subacromial bursa superiorly results in pain, tendon disintegration and functional deterioration as indicated by decreased joint mobility (Larsen et al., 1977; Lehtinen et al., 2001).

Hip joint involvement has also been documented in close to 15% of RA patients after only one year of disease onset. After 5 years, it was noted in one third of RA patients (Eberhardt et al., 1990). Hip joint involvement is difficult to appreciate owing to the challenges faced in assessing hip joint synovitis. In some RA cases there is rapidly progressive destruction of the joint. Femoral head destruction occurs in a specific sequence of events: depression, flattening, and loss (Eberhardt et al., 1990). The reason for rapid joint destruction in RA is yet to be elucidated, but some suggest the role of high levels of prescribed steroids resulting in a picture similar to avascular necrosis (Yun et al., 2012).

CONCLUSIONS

RA is a chronic inflammatory disease of the joints that results in pannus formation, joint swelling, erosions and deformity. Immune-pathogenic advances have helped to provide insight into RA disease mechanisms. However, much remains to be resolved. Although novel therapies have revolutionized the standard care of RA patients, patients should be tightly controlled in order to prevent the progression of the disease to debilitating and sometime life-threatening joint deformities.

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